

Hawai'i's Electricity Future: Solar Reform, Enhanced Geothermal, and the JERA LNG Proposal

Ethan Hartley Michael J. Roberts

University of Hawai'i at Mānoa — Economics, Sea Grant, UHERO

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All code, data, and results: github.com/mikejrob/oahu-electricity-future

Three decisions are on the table now

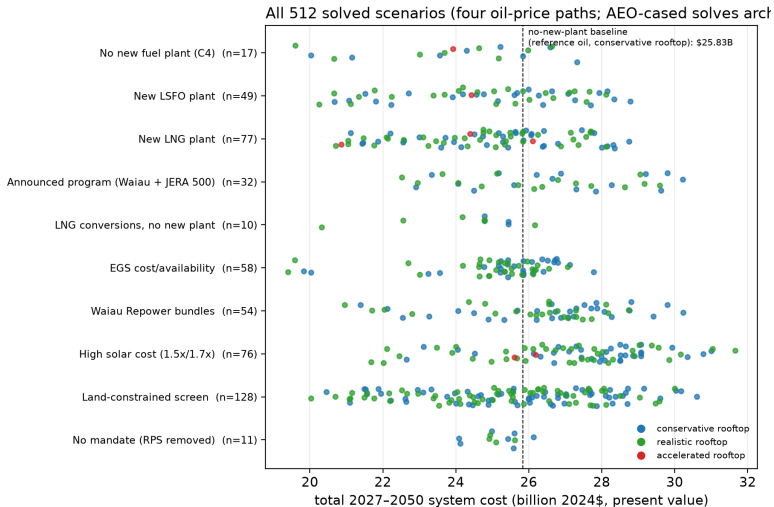
- **Waiau Repower** — six oil-fired turbines, approved March 2026
- **JERA proposal** — \$2B, 500 MW plant + LNG import terminal
- **Utility-scale solar** — procurement has stalled; land rules bind

O'ahu pays the nation's highest electricity prices. Each choice locks in decades.

What we built

- Open capacity-expansion model of O'ahu (Switch), 2027–2050
- Hourly operations: unit commitment, reserves, storage, EVs
- 500+ solved scenarios: oil prices, solar costs, land rules, LNG, geothermal
- Everything public — inputs, code, results, and an interactive explorer

The scenario space

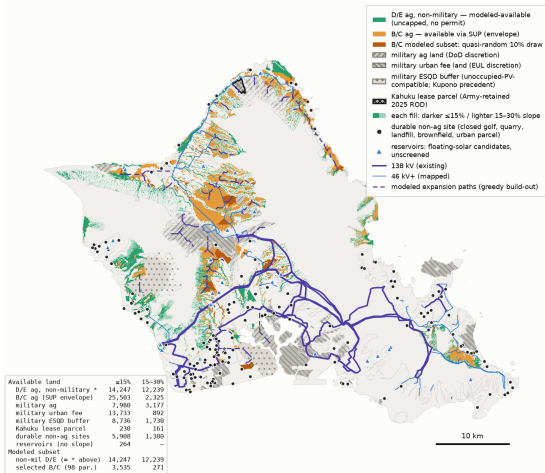


Solar and land

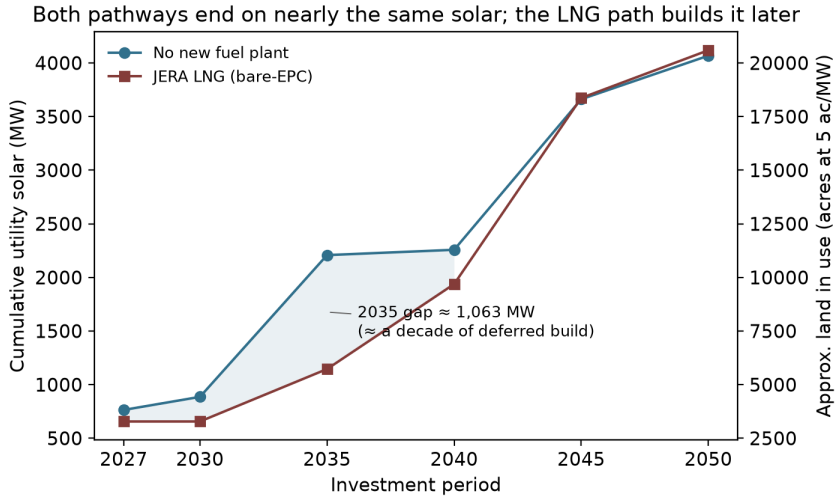
Plausibly available land, and what the model uses

Oahu: plausibly available solar land, the modeled subset, and transmission

Each fill has two shades: darker = $\leq 15\%$ slope, lighter = the 15–30% increment (10 m DEM). The modeled subset is non-military D/E ag (green) plus a quasi-random 10% B/C draw; military land is shown by tenure category (available at DoD discretion, not in the modeled subset). No grid-distance filter: published near-grid figures are smaller (notes/available-land-map.mdi).



How much land, and when



Rooftops change the land question

- Measured potential: about 4,000 MW on existing structures
- Three adoption paths: $\sim 1,000$ / 1,600 / 2,100 MW by 2050
- Fast rooftop growth: utility solar falls from 4,100 to 3,000 MW; land toward 15,000 acres
- What unlocks it: paying fair value for exports (Section 2.7)

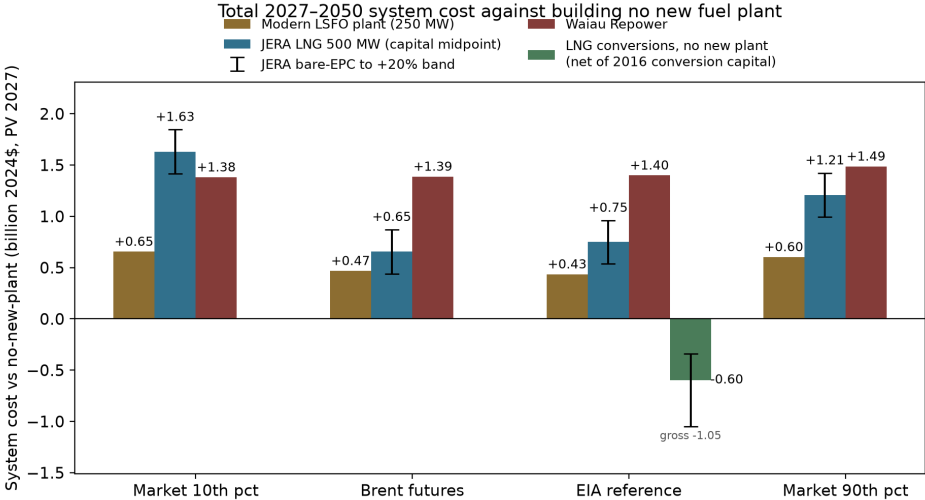
Hawai'i pays a large, unexplained solar premium

Utility solar here: 1.4 to 2.2+ $\times$ mainland cost

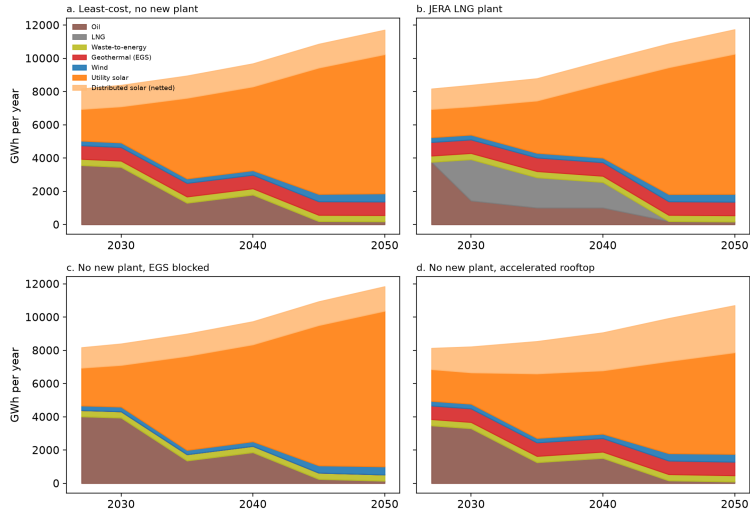
- Physical conditions explain only part
- 2018-era contracts (Phase 1, Kaua'i) came in *below* 1.8 $\times$
- Recent O'ahu repricings anchor the top of the range

Does a new fuel plant pay?

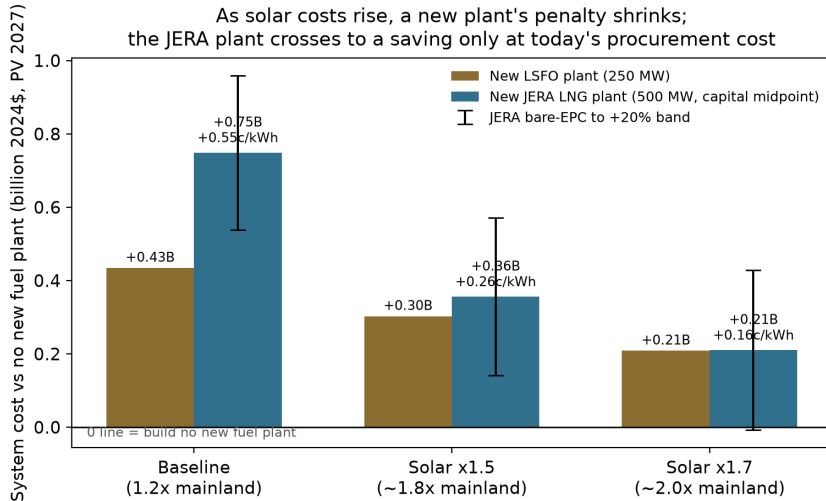
No proposed plant beats building no plant



What the system builds instead

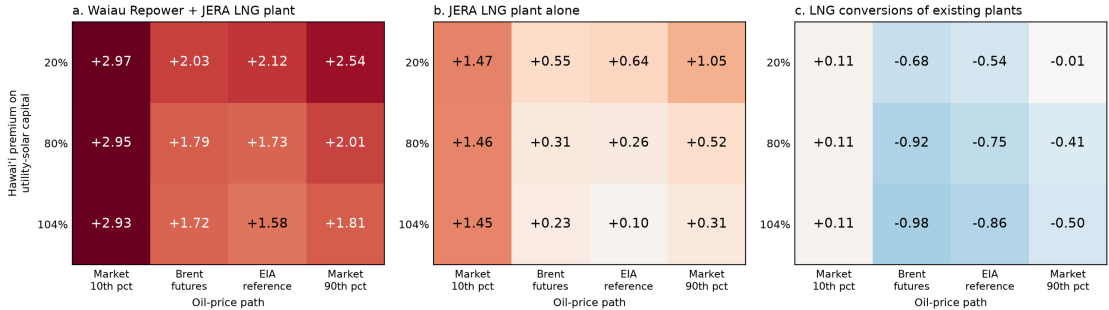


The verdict survives expensive solar



...and every oil price

Cost of each thermal commitment against building no new plant (billion 2024\$; red costs more, blue saves)



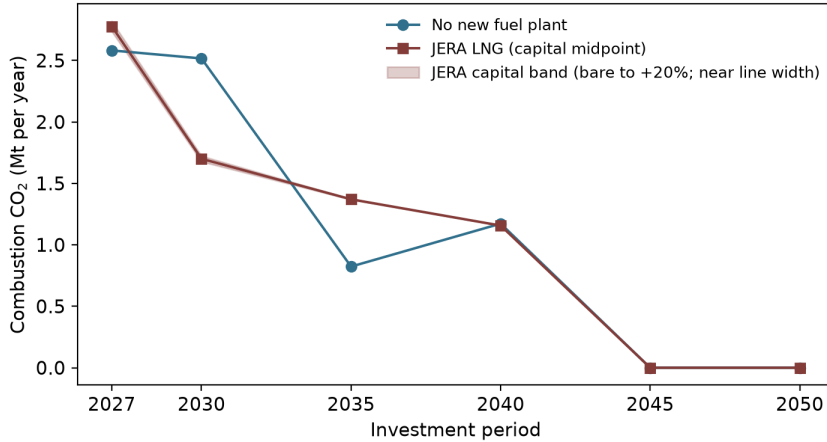
If LNG comes, existing plants beat new ones

LNG configuration (vs. no new plant)	Δ cost
New JERA plant (midpoint)	+\$0.7–1.6B
Convert Kalaeloa alone	−\$0.4B
+ Kahe 5 & 6, CIP turbine	−\$1.1B

Same fuel savings, no construction. Conversion budgets: ~\$1,800/kW before the case breaks.

LNG's climate case hangs on methane leakage

Annual combustion emissions: LNG cleaner around 2030, dirtier mid-2030s;
cumulative totals in Section 4.7



The leakage arithmetic

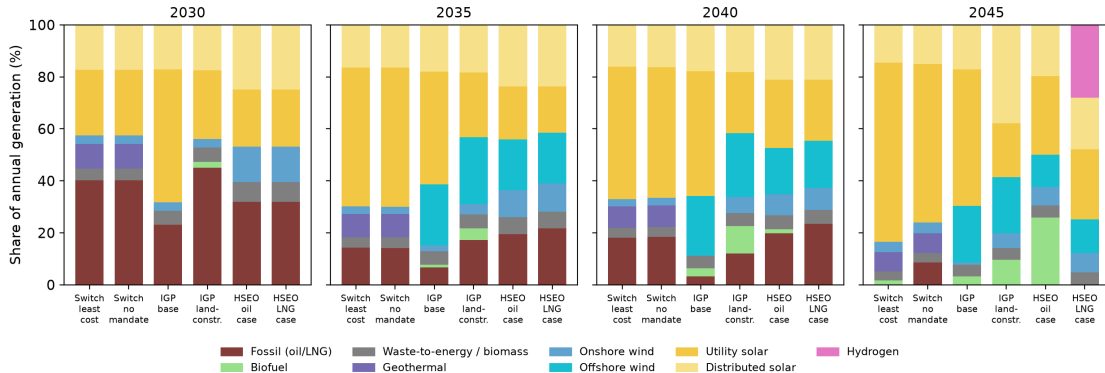
Greenhouse advantage gone above $\sim 0.9\%$
supply-chain leakage

- (100-year basis; $\sim 0.3\%$ on a 20-year basis)
- Measured U.S. supply-chain rates sit well above both thresholds

The plans on file

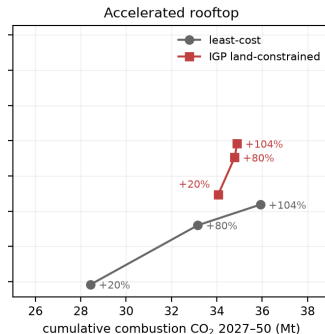
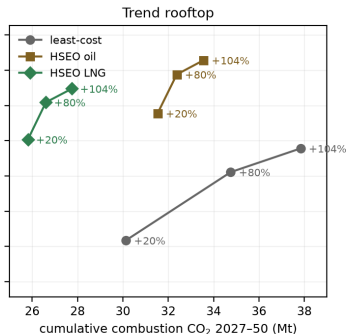
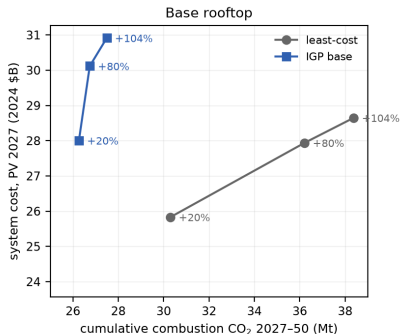
Our least-cost paths vs. HECO's IGP and HSEO's study

Generation mix by plan: this report's solved paths vs HECO's IGP and HSEO's Alternative Fuels Study



Every plan, priced on one ruler

Published plans vs the least-cost path as the solar premium rises: plans hold their mix (near-vertical); the optimum trades carbon for cost



Enhanced geothermal

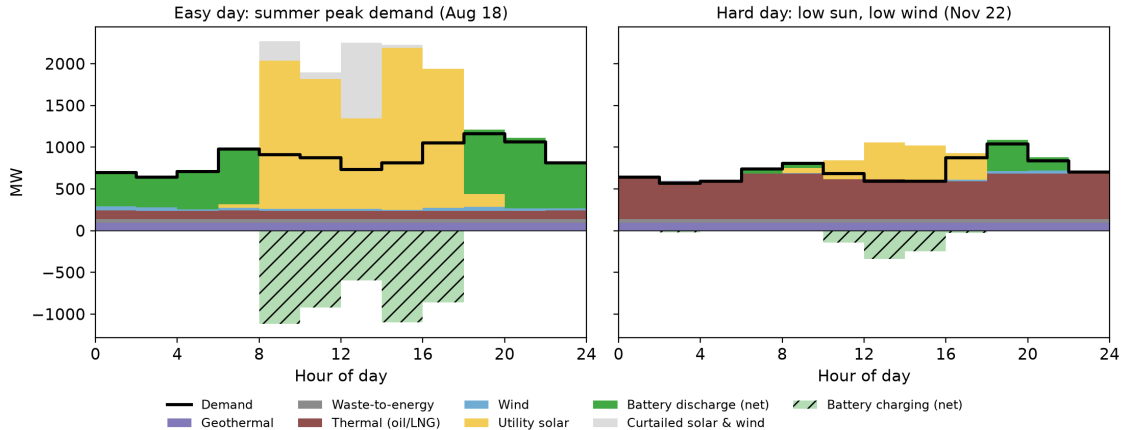
A small resource with outsized value

- First resource screen: ~ 100 MW identified on O'ahu
- The model builds all of it in the base case
- Blocking it costs $+\$0.6\text{B}$; at optimistic drilling costs the saving reaches $\sim \$1.0\text{B}$
- 100 MW displaces ~ 400 MW of solar land
- The resource at depth is unmeasured — could be far larger

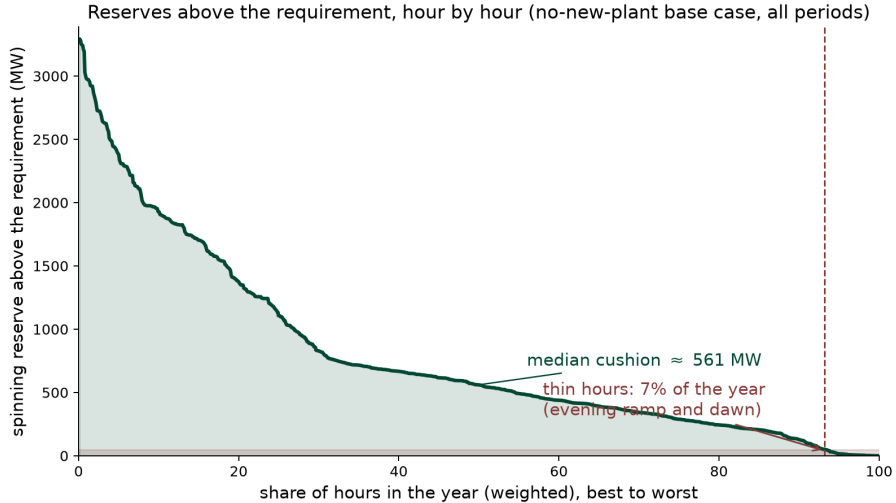
Reliability

The hard day is not the peak day

Hourly generation, no-new-plant base case, 2035



Resource adequacy: covered with room to spare



Resilience changes on a renewable grid

- Firm plants cover the daily-average net load (~ 610 MW in 2035); batteries carry energy to the evening peak ($\sim 1,270$ MW)
- Generation comes in thousands of pieces — the main risk becomes island-wide, multi-day, forecastable weather
- Two units out at a thin hour, at the filed 10% failure rate: ~ 6 h/yr — conventional standards run 2–8 h/yr
- The real tail: correlated failures and storage depth (a 16-day low-wind run is on record) — v2 tests both

The Waiau Repower fails its own cost test

- Adds +\$1.4B to system cost on every oil path
- Runs at 51% in 2030, under 32% by the late 2030s, ~0 from 2045
- A modern combined cycle delivers the same firm capacity at ~1/3 less per kW
- PUC cost cap: \$220–310M of the bill lands on shareholders

What reform unlocks

Three levers, no new technology required

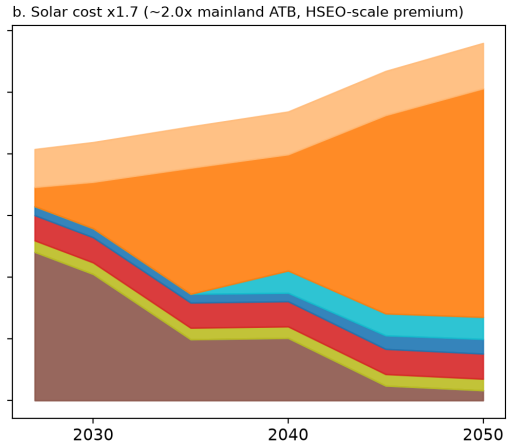
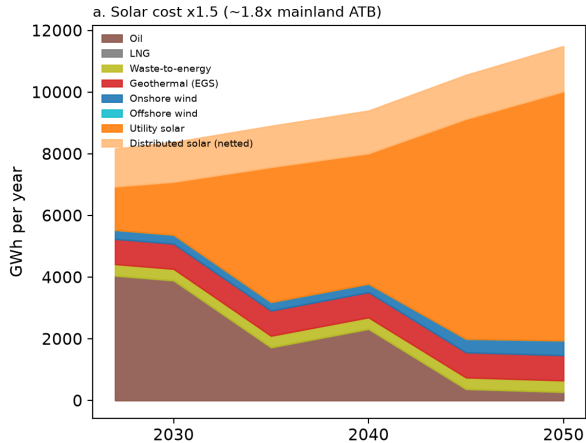
- **Land:** state siting rules exclude most of the island's flat, cleared, transmission-adjacent land
- **Procurement:** independent bid evaluation, standard contracts, published queues
- **Rooftops:** pay fair value for exported power

The savings at stake dwarf every supply-side proposal on the table.

The deeper reform: retail wheeling

- The Legislature has mandated retail wheeling — *how* is the open question
- Our proposal (Schedule RWS, in preparation): sellers and large buyers settle at **locational marginal prices**; the utility keeps the wires
- Ten-day provisional interconnection replaces the RFP gauntlet
- Real-time prices unlock demand flexibility (Imelda, Fripp & Roberts 2024, *AEJ: Policy*)
- Self-funding: a transition fund captures part of today's oil-priced margins

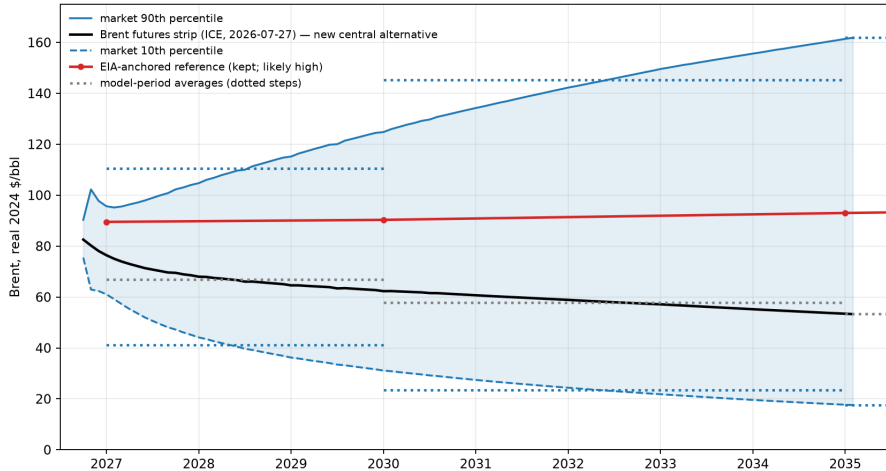
Even if solar stays expensive here, the answer holds



The oil bet

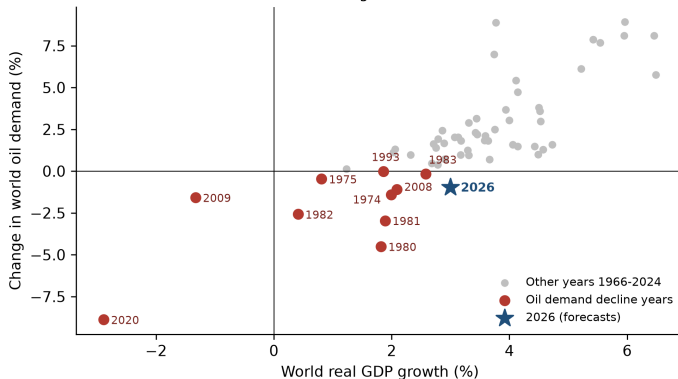
Four oil futures, priced from markets

Four oil-price cases: EIA reference, futures central, and the market 10th/90th band
(real 2024\$, deflated on TIPS breakevens; σ beyond liquid tenors = 0.30, documented)



World oil demand has stopped growing with GDP

Prior oil demand declines came with weak world economies.
2026's is forecast to occur near trend growth.



Sources: EI Statistical Review 2025 (oil, consumption basis, 1966-2024); World Bank WDI (GDP, market exchange rates); 2026: IEA OMR July 2026 (-1.0 mb/d forecast) and IMF WEO Update July 2026 (3.0% growth, PPP weights, ~0.5pp above the market-rate basis of the historical points; ~2.4% market-rate-comparable). 1993's dip reflects the FSU collapse.

Takeaways

Takeaways

- **No proposed fuel plant pays for itself** — not JERA's, not the Waiau Repower, under any oil or solar price we can defend
- **Solar + storage + existing plants** is the robust least-cost core; geothermal sweetens it
- **The binding constraints are rules, not resources** — land, procurement, and rooftop compensation

Kick the tires

`github.com/mikejrob/oahu-electricity-future`

- Full report, all inputs, all 500+ solved scenarios
- Interactive explorer — every figure, every configuration
- Comments to September 15, 2026; we answer publicly

Backup

System cost by trajectory and oil price (Table ES.1)

	Market 10th	Brent futures	EIA ref.	Market 90th
No new fuel plant	21.2	24.3	25.8	27.3
Modern LSFO plant	+0.7	+0.5	+0.4	+0.6
JERA LNG (midpoint)	+1.6	+0.7	+0.8	+1.2
Waiau Repower	+1.4	+1.4	+1.4	+1.5
Waiau + JERA	+3.1	+2.1	+2.2	+2.7

Billions of 2024\$, present value 2027–2050, 3% real.

Method guardrails

- 0.1% optimization tolerance; dominance checks across 698 scenario pairs gate every release
- 13 sample days per period include the record's hardest day (Nov. 22, 2008: low sun, weak trades)
- Reserves every hour: the largest unit online, plus regulation that grows with solar and wind output
- Conservative for renewables: no inter-day storage, no real-time pricing, distributed value uncredited

Since the draft: Kalaeloa signed for 30 more years

- New PPA signed July 27, 2026 (pending PUC approval)
- 208 MW firm, biofuel-capable repowering, ~2033–2063
- Capacity charge *fell* \$7/kW-yr — firm capacity without new construction keeps beating new plants